THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 75	Angle $= \lvert 30h - 5.5m \rvert$ (take the smaller one).
2	**(d)** 120	6 degrees per minute.
3	**(c)** 5	30° per hour.
4	**(a)** 160	Full turn is 360°.
5	**(c)** 27	A bisector divides the angle into two equal parts.
6	**(d)** 12	Ordered pairs of points.
7	**(c)** right	Find the angle first.
8	**(a)** 75	Straight angle is 180°.
9	**(b)** 10	Angle $= \lvert 30h - 5.5m \rvert$ (take the smaller one).
10	**(d)** 210	6 degrees per minute.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand $= 165^\circ$ from 12; minute hand $= 180^\circ$ (1); Smaller angle $= 15^\circ$ (1).	1+1
12	Bisector gives two equal angles (1); $\angle AOC = 2 \times 28 = 56^\circ$, type acute (1).	1+1
13	Hour hand: $4 \times 30 = 120^\circ$ (1); Minute hand: 4 full turns $= 1440^\circ$ (1).	1+1
14	Hour hand $= 80^\circ$ from 12; minute hand $= 240^\circ$ (1); Smaller angle $= 160^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Smaller angle $= 135^\circ$ (1); Type: obtuse (1); Larger angle $= 225^\circ$: reflex (1).	1+1+1
16	Anticlockwise total $= 60 + 70 = 130^\circ$ (1); Clockwise 40° is subtracted (1); Net turn $= 90^\circ$ anticlockwise (right) (1).	1+1+1
17	Total segments $= \frac{5 \times 4}{2} = 10$ (1); Segments having A as an end point: 4 (1); Segments without A: $10 - 4 = 6$ (1).	1+1+1
	OR: Smaller angle $= 20^\circ$ (1); Type: acute (1); Larger angle $= 340^\circ$: reflex (1).	1+1+1
18	Anticlockwise total $= 45 + 100 = 145^\circ$ (1); Clockwise 85° is subtracted (1); Net turn $= 60^\circ$ anticlockwise (acute) (1).	1+1+1
	OR: Total segments $= \frac{6 \times 5}{2} = 15$ (1); Segments having A as an end point: 5 (1); Segments without A: $15 - 5 = 10$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minute hand turns 240° (1); Hour hand turns 20° (1); Hour hand at 110° (1); Angle between hands $= 130^\circ$ (1); Type: obtuse (1).	1+1+1+1+1
	OR: $\angle AOD = 40 + 55 + 70 = 165^\circ$ (1); Type: obtuse (1); Remaining turn $= 360 - 165 = 195^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
20	Minute hand turns 120° (1); Hour hand turns 10° (1); Hour hand at 160° (1); Angle between hands $= 40^\circ$ (1); Type: acute (1).	1+1+1+1+1
	OR: $\angle AOD = 60 + 85 + 45 = 190^\circ$ (1); Type: reflex (1); Remaining turn $= 360 - 190 = 170^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	30° per hour.
2	**(b)** 120	Full turn is 360°.
3	**(a)** 38	A bisector divides the angle into two equal parts.
4	**(a)** 20	Ordered pairs of points.
5	**(c)** straight	Find the angle first.
6	**(c)** 90	Straight angle is 180°.
7	**(a)** 10	Angle $= \lvert 30h - 5.5m \rvert$ (take the smaller one).
8	**(a)** 240	6 degrees per minute.
9	**(c)** 9	30° per hour.
10	**(b)** 75	Full turn is 360°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bisector gives two equal angles (1); $\angle AOC = 2 \times 47 = 94^\circ$, type obtuse (1).	1+1
12	Hour hand: $5 \times 30 = 150^\circ$ (1); Minute hand: 5 full turns $= 1800^\circ$ (1).	1+1
13	Hour hand $= 250^\circ$ from 12; minute hand $= 120^\circ$ (1); Smaller angle $= 130^\circ$ (1).	1+1
14	Bisector gives two equal angles (1); $\angle AOC = 2 \times 65 = 130^\circ$, type obtuse (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Smaller angle $= 155^\circ$ (1); Type: obtuse (1); Larger angle $= 205^\circ$: reflex (1).	1+1+1
16	Anticlockwise total $= 90 + 120 = 210^\circ$ (1); Clockwise 30° is subtracted (1); Net turn $= 180^\circ$ anticlockwise (straight) (1).	1+1+1
17	Total segments $= \frac{7 \times 6}{2} = 21$ (1); Segments having A as an end point: 6 (1); Segments without A: $21 - 6 = 15$ (1).	1+1+1
	OR: Smaller angle $= 15^\circ$ (1); Type: acute (1); Larger angle $= 345^\circ$: reflex (1).	1+1+1
18	Anticlockwise total $= 35 + 75 = 110^\circ$ (1); Clockwise 50° is subtracted (1); Net turn $= 60^\circ$ anticlockwise (acute) (1).	1+1+1
	OR: Total segments $= \frac{8 \times 7}{2} = 28$ (1); Segments having A as an end point: 7 (1); Segments without A: $28 - 7 = 21$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minute hand turns 300° (1); Hour hand turns 25° (1); Hour hand at 235° (1); Angle between hands $= 65^\circ$ (1); Type: acute (1).	1+1+1+1+1
	OR: $\angle AOD = 35 + 90 + 120 = 245^\circ$ (1); Type: reflex (1); Remaining turn $= 360 - 245 = 115^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
20	Minute hand turns 240° (1); Hour hand turns 20° (1); Hour hand at 290° (1); Angle between hands $= 50^\circ$ (1); Type: acute (1).	1+1+1+1+1
	OR: $\angle AOD = 75 + 80 + 95 = 250^\circ$ (1); Type: reflex (1); Remaining turn $= 360 - 250 = 110^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 44	A bisector divides the angle into two equal parts.
2	**(d)** 30	Ordered pairs of points.
3	**(b)** acute	Find the angle first.
4	**(d)** 60	Straight angle is 180°.
5	**(a)** 105	Angle $= \lvert 30h - 5.5m \rvert$ (take the smaller one).
6	**(d)** 300	6 degrees per minute.
7	**(d)** 11	30° per hour.
8	**(a)** 50	Full turn is 360°.
9	**(b)** 56	A bisector divides the angle into two equal parts.
10	**(c)** 42	Ordered pairs of points.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand: $6 \times 30 = 180^\circ$ (1); Minute hand: 6 full turns $= 2160^\circ$ (1).	1+1
12	Hour hand $= 345^\circ$ from 12; minute hand $= 180^\circ$ (1); Smaller angle $= 165^\circ$ (1).	1+1
13	Bisector gives two equal angles (1); $\angle AOC = 2 \times 95 = 190^\circ$, type reflex (1).	1+1
14	Hour hand: $3 \times 30 = 90^\circ$ (1); Minute hand: 3 full turns $= 1080^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Smaller angle $= 145^\circ$ (1); Type: obtuse (1); Larger angle $= 215^\circ$: reflex (1).	1+1+1
16	Anticlockwise total $= 110 + 80 = 190^\circ$ (1); Clockwise 65° is subtracted (1); Net turn $= 125^\circ$ anticlockwise (obtuse) (1).	1+1+1
17	Total segments $= \frac{9 \times 8}{2} = 36$ (1); Segments having A as an end point: 8 (1); Segments without A: $36 - 8 = 28$ (1).	1+1+1
	OR: Smaller angle $= 25^\circ$ (1); Type: acute (1); Larger angle $= 335^\circ$: reflex (1).	1+1+1
18	Anticlockwise total $= 25 + 125 = 150^\circ$ (1); Clockwise 70° is subtracted (1); Net turn $= 80^\circ$ anticlockwise (acute) (1).	1+1+1
	OR: Total segments $= \frac{10 \times 9}{2} = 45$ (1); Segments having A as an end point: 9 (1); Segments without A: $45 - 9 = 36$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minute hand turns 300° (1); Hour hand turns 25° (1); Hour hand at 85° (1); Angle between hands $= 145^\circ$ (1); Type: obtuse (1).	1+1+1+1+1
	OR: $\angle AOD = 50 + 65 + 110 = 225^\circ$ (1); Type: reflex (1); Remaining turn $= 360 - 225 = 135^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
20	Minute hand turns 240° (1); Hour hand turns 20° (1); Hour hand at 320° (1); Angle between hands $= 80^\circ$ (1); Type: acute (1).	1+1+1+1+1
	OR: $\angle AOD = 85 + 70 + 55 = 210^\circ$ (1); Type: reflex (1); Remaining turn $= 360 - 210 = 150^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.